

Statement of Purpose of Hrishikesh Belagali

My research interests are centered around a single question: how can we model complex physical systems effectively? I always thought science was about understanding the natural world through observation and experimentation. My research experience has revealed that computation is an equally powerful tool, which can provide unique insights inaccessible to traditional experiments. My interest in answering this question has guided me across multiple computational frameworks, including molecular simulation, quantum algorithms, and machine learning techniques. By participating in the Science Undergraduate Laboratory Internship, I aim to deepen my understanding of quantum simulation and optimization methods by learning from experts in the field. This program will help me explore the field of quantum algorithms for molecular simulation and understand how such methods can address computational bottlenecks, which will be invaluable as I consider PhD programs in quantum computing.

As an undergraduate research assistant in the Vermaas Lab, I work on molecular dynamics simulations of biomolecular systems using VMD and NAMD. Over the past year, I have been working on characterizing water transport mechanisms in aquaporin channels in plants, which are vital for maintaining cellular water balance under drought stress. Understanding this mechanism is essential for bioengineering efforts to improve plant drought tolerance. During this study, I faced challenges quantifying the transport mechanism because of the large system size, which restricted the simulation length and sampling. I explored alternative methods for simulation such as grid-steered molecular dynamics, which taught me to consider computational constraints while studying complicated systems.

My experience with the aquaporin project highlighted the limitations of classical simulation. Simulating larger systems, capturing statistically rare events, and reaching biologically significant timescales is computationally expensive, which limits the scope of phenomena we can study. These limitations led me to explore alternative computational approaches, of which quantum computing stood out to me because it leverages principles of quantum mechanics to solve complex problems. In order to explore my interest in quantum computing, I work as an Honors Research Scholar under Professor Ryan LaRose with a focus on generalized quantum simulators. My research involves studying group-theoretic methods and compilation optimization techniques in Python and Cirq to develop methods for simulating quantum algorithms on classical computers. The development of advanced quantum simulation techniques is useful for benchmarking and refining larger-scale quantum algorithms, which might not be feasible to run on current quantum hardware. By contributing to this research, I've learned to consider both theoretical performance and practical challenges of implementing quantum algorithms.

Alongside formal research, I have built several computational physics projects using Python, NumPy, Matplotlib, and Qiskit. In my first semester of college, I was exposed to Monte Carlo algorithms, and was amazed by how directed random search could be used to solve a variety of difficult problems. I implemented a simulated annealing algorithm for Quadratic Unconstrained Binary Optimization (QUBO) problems and later wrote a quantum optimization algorithm using adiabatic evolution in Qiskit. Experimenting with different evolution schedules and total evolution times helped me understand the limitations of current quantum simulators and improved my scientific programming skills.

I am eager to further explore the intersection of molecular simulation and quantum algorithms through the SULI program at Oak Ridge National Laboratory. ORNL's focus on biophysics and simulations, the Quantum Science Center's QHPC initiative and research on hybrid quantum-classical algorithms closely align with my interests. I am particularly excited about the prospect of working under the guidance of Dr. Ryan Bennink, whose research on quantum high performance computing is very intriguing. ORNL's interdisciplinary research environment will allow me to improve my scientific communication and collaboration skills, which will be beneficial for my career. I believe that my intersection of molecular dynamics and quantum algorithms research experience will allow me to meaningfully contribute to ongoing projects at ORNL, and the skills that I gain through this program will prepare me for graduate school and a career in research. In the long term, I'm interested in working on methods that bridge molecular simulation and quantum algorithms to model materials and biomolecular systems effectively.